

Instrument/Finger Thoracostomy

**** REQUIRES SPECIFIC CREDENTIALING ****

Instrument/Finger Thoracostomy is a procedure in which an incision is made into the chest wall and pleural cavity to drain air or fluid. This procedure may be done either alone or in conjunction with needle decompression of the chest.

Indications for Instrument/finger thoracostomy

- Tension pneumothorax not relieved with needle thoracostomy
- Suspected hemothorax causing hypotension or inability to ventilate
- Traumatic cardiac arrest

Contraindications

- No absolute contraindications for emergent thoracostomy but patient should be in extremis with shock, hypoxia or inability to ventilate – unrelated to airway management
- Use caution to prevent provider finger laceration, when patient has multiple rib fractures

Relative Contraindications

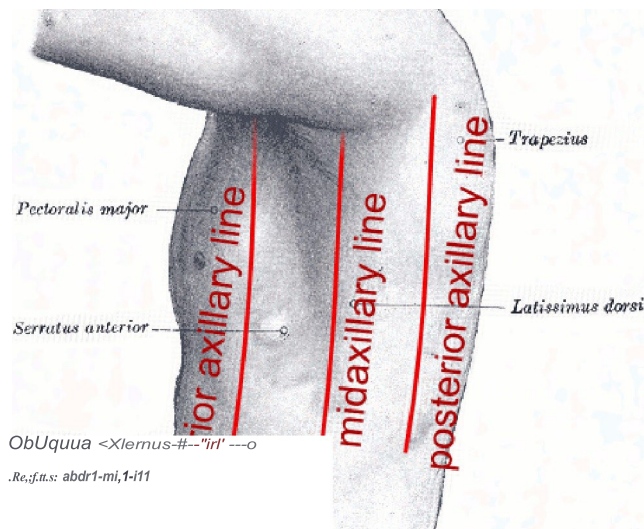
- Broken ribs
- Overlying skin infection
- Anticoagulation or bleeding disorder

Equipment needed for finger thoracostomy

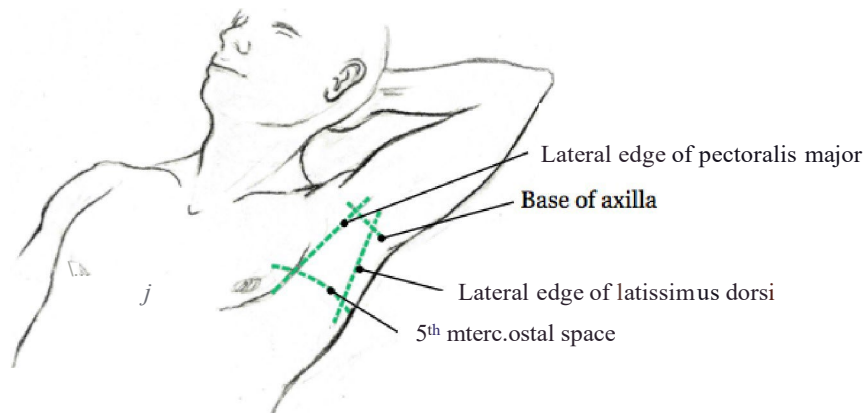
- Scalpel #10 blade (or whatever is available)
- Finger OR
- Kelly clamp (instrument)

Anatomy for finger thoracostomy

- Incision will be made at the 5th intercostal space, between the anterior axillary and midaxillary line (see image)



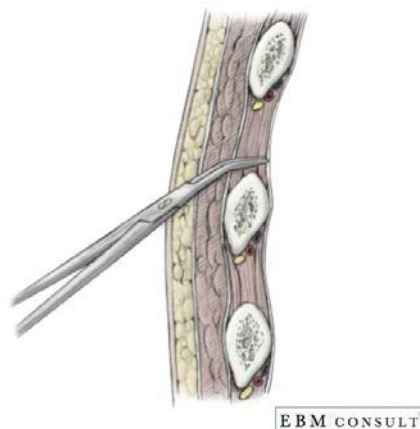
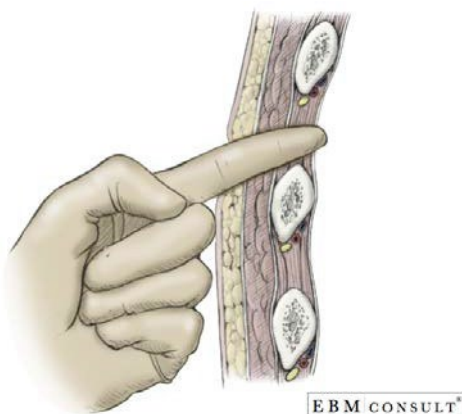
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Instrument/Finger Thoracostomy Procedure

- 1) Prepare equipment.
- 2) Place patient into supine position with ipsilateral arm abducted and externally rotated (so that hand is under or above the head).
- 3) Clean area with sterilizing prep, time permitting.
- 4) Identify the 5th intercostal space, between the anterior axillary and midaxillary line.
- 5) Using a #10 blade (or available scalpel), make a 3-4 cm transverse incision on the fifth rib – in the direction from the anterior axillary line to the midaxillary line.
- 6) Using a finger (or Kelly clamp if available) bluntly dissect down to the 5th rib and 4th intercostal space.
- 7) Using a finger (or Kelly clamp) push your finger (or clamp) over the top of the 5th rib, into the 4th intercostal space into the pleural cavity. You will feel a definitive pop once you have entered the space and there may be a rush of air or fluid from the cavity.
- 8) If a finger is used, confirm correct position and sweep between the lung and pleural wall to remove any adhesions. Take care not to damage the lung during this process.



- 9) If a Kelly clamp is used, place your index finger over the curve of the clamp to stabilize and prevent the clamp from damaging the lung. Once the clamp is in the pleural cavity, open the clamp to increase the incision. Then insert a finger to confirm correct position and sweep between the lung and pleural wall to remove any adhesions. Take care not to damage the lung.